SOFTWARE CHANGE IMPLEMENTATION REPORT

CR-020: Student able change their password

Field	Details
Change Request ID	CR-020
Change Request Title	Student able change their password
Student Name	P.Tharanan
Student ID	JF/ICT/24/20
Implementation Status	Completed
Date	14 October 2026

1. Change Request Summary

1.1 Business Requirement

Currently, student passwords can only be assigned by the system administrator during the student registration process. To improve security and user autonomy, the system must introduce a new feature in the student dashboard allowing students to securely change their own passwords.

Before implementation:

- Student password cannot changeable.
- System administrator can add password during the student registration.
- No change password currently available in the student dashboard

1.2 Requested Change

Add a change password in the student dashboard.

The implementation must:

- Student Change their password.
- Password changing time Must login.
- Password, Re-enter password field are in the change password section.
- In password field value must match Re-enter password field.

2. Implementation

2.1 Implementation Summary

CR-020 adds a password-change feature to the student dashboard My Profile section. A signed-in student can submit their current password, a new password, and confirmation of the new password.

The student dashboard provides a Change password action and a three-field form. The authenticated POST /api/auth/change-password endpoint verifies the current password with bcryptjs, requires the new password and confirmation to match, then hashes the new password with 10 salt rounds before updating the authenticated user's existing users.password_hash value. No schema change was required.

2.2 Affected Files

File	Change	Purpose
`backend/src/controllers/authController.js`	Modified	Verify the current password and hash/update the new password.
`backend/src/models/userModel.js`	Modified	Retrieve the authenticated user's hash and update it by user_id.
`backend/src/routes/authRoutes.js`	Modified	Register the authenticated change-password endpoint.
`frontend/src/services/authService.js`	Modified	Submit password-change values to the authenticated API.
`frontend/src/pages/StudentDashboardPage.jsx`	Modified	Add the profile action, three-field form, and feedback states.
`frontend/src/styles/studentDashboard.css`	Modified	Style the password form and profile security action using existing dashboard conventions.

Additional files: None

2.3 Technical Changes

Password Storage

No database schema migration was required; the existing users.password_hash column stores the password hash.
The password-change controller uses bcryptjs to hash the new password with 10 salt rounds before storage.
The new endpoint returns a success message only; it does not return the password or password hash.
The password update uses a parameterized query scoped to the authenticated user's user_id.
Authenticated Password Change
POST /api/auth/change-password is protected by the existing authenticate middleware.

The target user_id is taken from the verified JWT rather than request data.
The controller requires oldPassword, newPassword, and confirmPassword.
The backend rejects new-password values that do not match their confirmation.
bcrypt.compare checks the submitted old password against the stored password_hash.
The controller hashes and stores the new password only after the old password matches.
The existing login endpoint continues to verify credentials using bcrypt.compare.
No database schema or student CRUD route changes were needed.
Student Dashboard

My Profile now includes a Change password action under Account security.
Selecting the action opens fields for the old password, new password, and re-entered new password.
The dashboard displays password errors and a success message, and clears the fields after a successful update.
Implementation Safeguards

The student dashboard remains behind the STUDENT protected route, and the password endpoint requires a valid authentication token.
An incorrect old password returns before any password update is performed.
The database update is parameterized and scoped to the user identified by the verified token.

2.4 Implementation Logic

The student signs in and opens My Profile in the student dashboard.
The student selects Change password and submits the old, new, and confirmation values.
The dashboard and backend reject missing fields or a new password that does not match its confirmation.
The server retrieves the authenticated user's stored hash and verifies the old password with bcrypt.compare.
After successful verification, the server hashes the new password with bcryptjs (10 salt rounds) and updates users.password_hash.
The dashboard clears the submitted values and shows success after the API update; errors are displayed to the student.
The existing users.password_hash column is reused; no migration is required.
The updated hash is used by the existing login flow for subsequent sign-ins.
Existing student profile and CRUD routes were not changed.

2.5 Error and Data Handling

Missing fields, mismatched new passwords, and an incorrect old password return an HTTP 400 response.
Missing, invalid, or expired authentication tokens are rejected by the existing middleware.
Database errors return a generic server error; password values are not included in API responses.

3. Testing & Verification

3.1 Testing Approach

Validation for this report update included focused frontend ESLint, backend Node.js syntax checks, a frontend production build, and source review. The four supplied screenshots show the profile, password form, incorrect-current-password response, and success state. Database-backed password-change integration and CRUD regression were not run.
Verification methods:
Four supplied UI screenshots: profile, change-password form, incorrect old password, and successful update (image files still need embedding).
Focused ESLint on StudentDashboardPage.jsx and authService.js
Node.js syntax checks on authController.js, userModel.js, and authRoutes.js
Vite frontend production build
Source review of authentication, password verification, and hashed database update
Database-backed password-change and subsequent-login verification: not run
Student CRUD regression: not run
Environment: Development
Browser: User-supplied reference screenshots; Test Role: Student; live API/database interaction not re-run

3.2 Test Cases

Test ID	Test Case	Expected Result	Actual Result	Status
TC01	View student My Profile	Profile shows Change password under Account security	Supplied screenshot shows the profile action	Shown
TC02	Open password form	Form shows old, new, and re-enter password fields	Supplied screenshot shows all three fields	Shown
TC03	Submit incorrect old password	Password remains unchanged and an error is shown	Supplied screenshot shows the incorrect-password message	Shown
TC04	Submit valid password change	Password is updated and success feedback is shown	Supplied screenshot shows the success message; database update not independently exercised	Shown
TC05	Submit mismatched new passwords	Request is rejected before the database update	Frontend and backend validation paths reviewed	Reviewed
TC06	Call endpoint without valid authentication	Request is rejected; update is scoped to token user_id	Route middleware and controller source reviewed	Reviewed
TC07	Verify and store password	Old hash is compared; new password is bcrypt-hashed before update	Controller and user model source reviewed	Reviewed
TC08	Run static checks and build	Lint, backend syntax checks, and frontend build pass	Focused ESLint, node --check, and Vite build passed	Pass
TC09	Database integration and CRUD regression	Password change works against MySQL; existing CRUD remains operational	Not run during this report update	Not run

3.3 Test Summary

Test Result	Count
Total Verification Checks	9
Shown in supplied screenshots	4
Verified by source review	3
Static checks/build passed / DB & CRUD not run	1 / 1

The supplied UI screenshots illustrate the four requested states; focused ESLint, backend syntax checks, and the frontend production build passed. Database-backed password-change verification and CRUD regression were not run. The screenshot image files still need to be embedded in this report.

4. Acceptance Criteria Verification

ID	Acceptance Criterion	Verification	Status
AC-001	Student can access Change password from My Profile.	Supplied profile screenshot	Met
AC-002	Change-password form includes old, new, and confirmation fields.	Supplied form screenshot	Met
AC-003	Current password is checked before an update.	bcrypt comparison source review	Met
AC-004	New password and confirmation must match.	Frontend and backend validation source review	Met
AC-005	New password is stored as a hash, not plain text.	bcrypt hash and parameterized update source review	Met
AC-006	Incorrect current password is rejected without changing the password.	Supplied error screenshot and early-return source review	Met
AC-007	Successful change displays confirmation to the student.	Supplied success screenshot	Met
AC-008	Existing login and student CRUD routes remain in place.	Source review; database regression not run	Met

5. Evidence

Four screenshots are referenced below: student profile, change-password form, incorrect old password, and successful update. Their image files were not available in the workspace to embed in this document.

5.1 Evidence 01 - Student My Profile

Related: AC-001

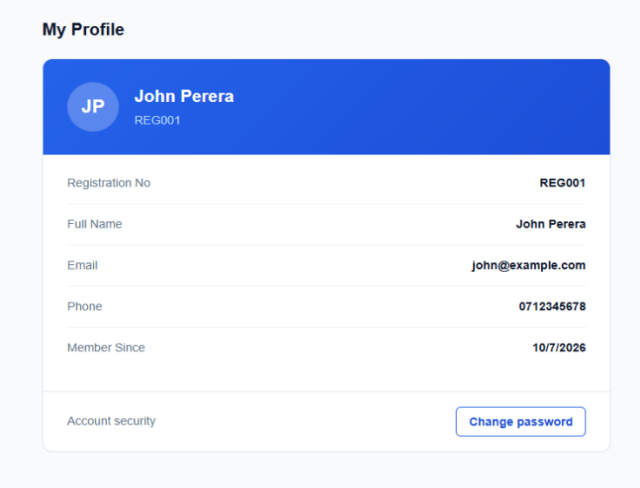


Figure 1: Student dashboard My Profile showing the Change password action.

Evidence 02 - Change Password Form

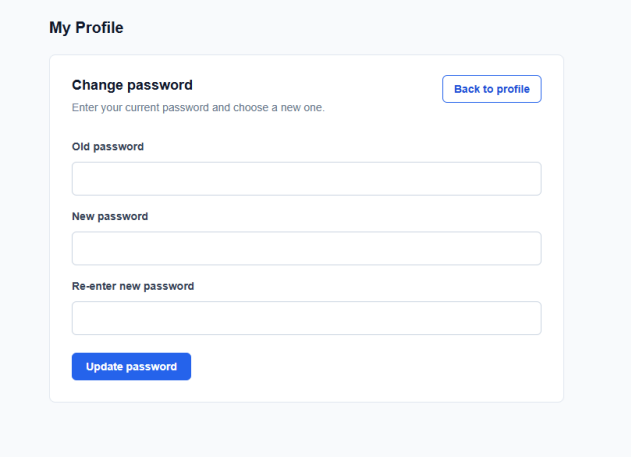


Figure 2: Change-password form with old password, new password, and re-enter new password fields.

Evidence 03 - Incorrect Current Password

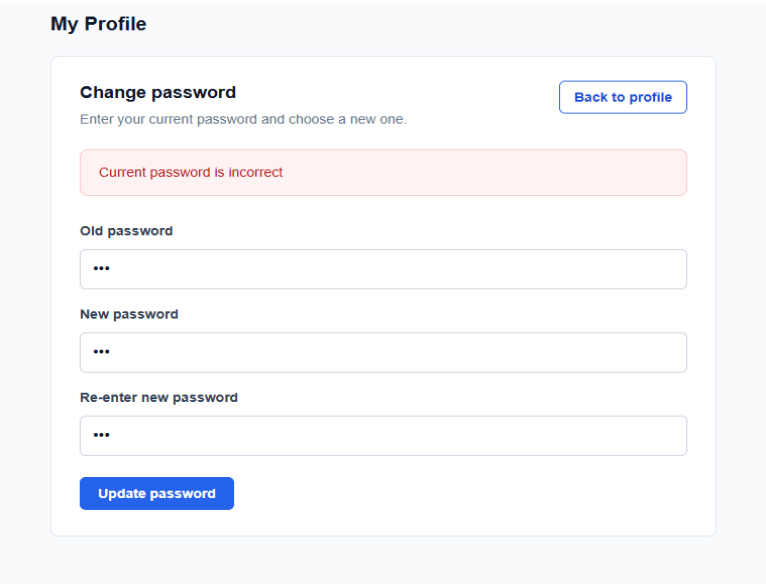


Figure 3: Incorrect current password is rejected and an error message is displayed.

Evidence 04 - Successful Password Change

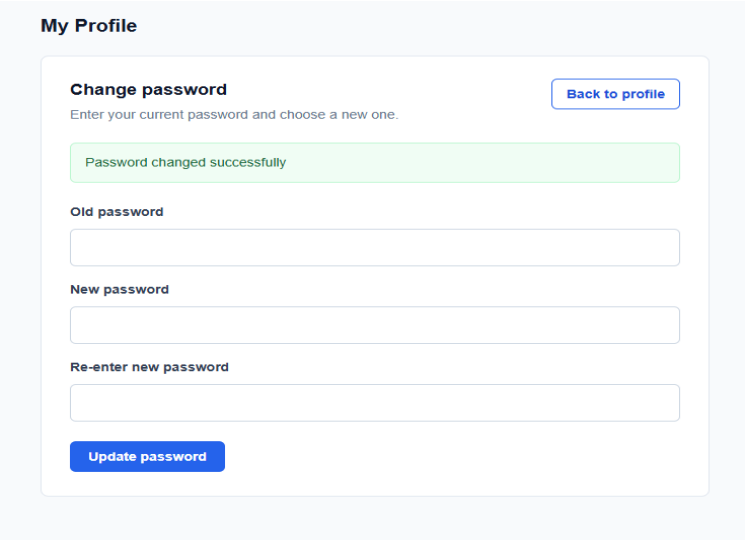


Figure 4: Successful password change is confirmed to the student.

6. Final Result

Item	Result
Requirements	Met by implementation and supplied UI evidence
Implementation	Completed
Testing	Lint, syntax checks, and frontend build passed; DB integration not run
Acceptance Criteria	Met by source review and supplied UI evidence
Existing CRUD	Unchanged; regression not run
Final Status	Implemented; database verification and screenshot embedding remain outstanding

6.1 Final Summary

CR-020 adds a student-facing password-change form to My Profile and an authenticated API endpoint. The backend verifies the current password against its stored bcrypt hash, checks that the new password matches its confirmation, and stores only a bcrypt hash for the authenticated account. No database schema change was required.

Database-backed password-change verification and CRUD regression were not run. The four screenshot image files still need to be embedded in the Evidence section.

Other implementation issues:
Database-backed API verification and embedding the four supplied screenshots remain outstanding.

Student Name: ___P.Tharanan____

Student ID: JF/ICT/24/20

Date: 14 October 2026

Signature: _____P.Tharanan_____

~ END OF REPORT ~